SE Journals in 2036: Looking Back at the Future We Need to Have

Tim Menzies[1], Paris Avgeriou[2], Robert Feldt[3], Mauro Pezzè[4],
Abhik Roychoudhury[5], Miroslaw Staron[6], Sebastian Uchitel[7], Thomas Zimmermann[8]
[1]NC State University, USA [2]University of Groningen, the Netherlands [3]Chalmers University, Sweden [4]USI, Switzerland
[5]National University of Singapore [6]Univesity of Gothenburg, Sweden [7]Imperial College, UK [8]UC Irvine, USA
timm@ieee.org,p.avgeriou@rug.nl,robert.feldt@chalmers.se,mauro.pezze@usi.ch
abhik@nus.edu.sg,miroslaw.staron@chalmers.se,sebastian.uchitel@gmail.com,tzimmer@uci.edu

Abstract

In 2025, SE publishing faces an existential crisis of scalability. As our communities swell globally and integrate fast-moving methodologies like LLMs, traditional peer-review practices are collapsing under the strain. The "bureaucratic anomaly" of monolithic review has become mathematically unsustainable, creating a stochastic "lottery" that punishes novelty and exhausts researchers.

This paper, written from the perspective of 2036, documents potential solutions. Here, the editors of ASE, EMSE, IST, JSS, TOSEM and TSE dream a collective dream of a brighter future. In summary first we stopped fighting (The Journal Alliance). Then we fixed the process (The Lottery / Unbundling / Fixing the Benchmark Graveyard). And then we fixed the culture (Cathedrals/Bazaars).

1 Houston, We have a Problem

Researchers can be surprised by the differences between the craft of research and the mechanics of publication.

- To the individual, research is often solitary (or small team) task, that is a quality-driven pursuit—an intense scrutiny of specific ideas where success is measured by depth of insight and/or breadth or impact.
- Publication, on the other hand, is a complex process defined, at least to some extent, by scale. Experiments at AI conferences [1] reveal that independent review teams disagree on 50% of accepted papers. This suggests that peer review and "getting published" are stochastic rather than a precise measure. Our own analysis (shown in Figure 1) showed that just like AI, SE reviewing is more stochastic-based than merit-based.

Speaking as the editors of some high volume SE journals (ASE, EMSE, IST, JSS, TOSEM, TSE) we view this landscape through the lens of Price's Law [7]; i.e. 50% of scientific contributions are generated by the square root of the research population ($\sqrt{N}$). This creates a unique logistical imperative: we need methodologies to rapidly sample the total population (N) to identify those core contributors.

But our duty extends beyond simply harvesting the "breakthroughs." A healthy journal ecosystem must also support the foundational trial-and-error—the "negative" results and iterative attempts—that allow the high-impact work to emerge. We must efficiently filter for the signal, without discarding the necessary noise of the scientific process.

A 2025 editorial in TOSEM [10] underscored the large volume of submissions. In terms of concrete numbers, there were around 2000 submissions to TOSEM in 2025 including both first submissions and revisions. These submissions were handled within 67 days on average, which does put a high load on the entire reviewer community. Associate editors can send up to 20 invitations to secure reviewers. These bottlenecks can cause delays for certain papers. JSS decided in 2025 to establish a cap of 20 review invitations per submission. If this limit was reached, the submission was assumed to not fit the scope of the journal, and was desk-rejected. While this draconian measure may seem unfair, it is one way to address wasting too much time for the authors and editors.

When researchers and practitioners turn to arXiv for real-time advances in LLMs, traditional forums are being left behind, rendered irrelevant. The community attempted to patch these leaks:

- To ensure that good papers at journals were not ignored, the Journal First Initiative[1] allowed authors of published journal papers to present those at major conferences.
- To tap the wealth of research that may be languishing in arxiv, the Journal Ahead Workshop (JAWs[2] was organized initially by TOSEM and TSE for the International Conference on Software Engineering (ICSE) 2026 (and then later, by other journals at other venues). The aim of the JAWs workshop was to invite early ideas to be presented in a workshop of a major conference, subsequent to which selected papers may be submitted to a journal.

While timely, these initiatives were not enough. They did not address the core scalability problem, nor the friction caused by forcing fast-moving "Bazaar" artifacts into slow "Cathedral" review cycles. To achieve true scalability, we needed to adopt an engineering perspective on the review process itself. What was its true function? Where, precisely, were we wasting resources?

Taking that first step required us to confront the central myth of our discipline: that peer review is a reliable filter through which specialists objectively identify quality. For a brief period, the community operated under the somewhat misguided assumption that the only path to scientific truth lay through a Byzantine architecture of rebuttals, revisions, and gate-keeping. By 2026 it was very clear that that kind of peer review had become mathematically unsustainable and empirically suspect.

We say this because, in any active research area, it is mathematically impossible for reviews from N authors to be conducted by the $\sqrt{N}$ most active researchers.

- Table 1 shows the number of authors of papers with "top 10 citations" at leading SE venues from 2013 to 2023 (from [6]).
- The table compares top-10 authors to a "back-of-the-envelope" calculation for all the other authors at those venues. In this

Submitted to the Future of SE track, ICSE 2026.

[1]Initially organized in 2016 by Matt Dwyer.
[2]Suggested by Mauro Pezze in 2025 and then conducted in 2026 jointly with TOSEM

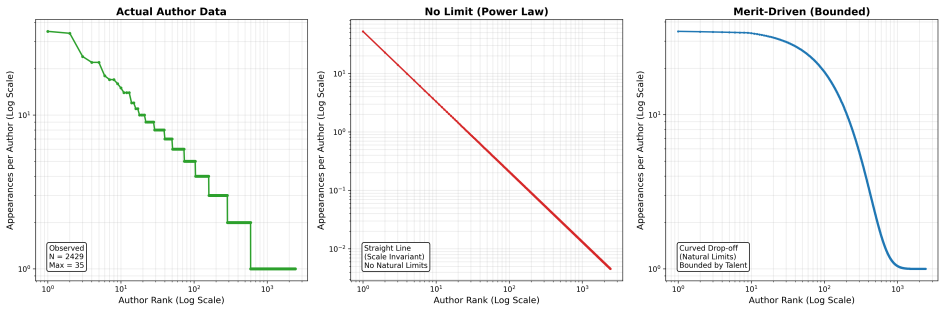

Figure 1: Consider two citation models. (a) Stochastic: Feedback loops cause unbounded citations; i.e. authors with many citations always earn yet more citations due to their viability. This would appear as a straight line on log-log graph of "paper rank" versus "citations standing". (b) Merit-based: Quality has some natural limits where most author quality clusters around a mean, with a few large outliers. Citations from such a population should appear like a bell curve, which on a log-log graph would be a non-linearity. The Green plot (left) shows data on authors whose papers scored ten most citations in leading SE forums 2013 to 2023 [6]. Note that the SE data matches a power law distribution (center) better than a merit-driven curve (right).

Group	N	Cites	Assumptions
Top-10 Authors	450–550	150–1000+	μ = 6–10 papers/author
Other Authors	~18,000	5–20	μ = 2.5 papers/author

Table 1: SE has a small core of most cited researchers.

sample, the SE field has $\approx 19,000$ researchers publishing at senior venues, of which $\frac{500}{19,000} \approx 2.6\%$ are the most active *core*[3].

This skew creates a critical asymmetry: the core cannot possibly review the massive volume of the "long tail," turning the reviewer shortage into one of scale rather than quality. When the core does not review the rest of the field, potentially useful work is overlooked, and reviewing becomes a random process rather than a method for promoting new groundbreaking ideas with a short review turnaround time. There have been concerns as well about allocating reviews based on submissions. Since certain authors are very prolific - allocating reviews based on submission may cause certain authors to have a huge say in the paper acceptance. This remains a consideration.

This paper outlines how we escaped that randomness. Writing from the perspective of 2036, we document the "Great Restructuring" of the last decade. Faced with the collapse of the traditional model, the community stopped trying to patch the leak and instead redesigned the pipe. What follows are the six pivotal reforms—ranging from political alliances to automated infrastructure—that transitioned software engineering from a cottage industry into a scalable scientific discipline.

2 Six Fixes for SE Journals

In summary first we stopped fighting (The Journal Alliance). Then we fixed the process (The Lottery / Unbundling / Fixing the Benchmark Graveyard). And then we fixed the culture (Cathedrals/Bazaars).

[3]This number is larger than what might be predicted by Price's law [7] since $\sqrt{18,500} = 136 < 500$. Nevertheless, it is still consistent with "most of the highly recognized work comes from a very small core".

2.1 The SE Journal Alliance

Before we could fix the mechanics of peer review, we had to fix the structure of our venues.

By 2025, the competitive dynamic between venues had created increasing inefficiencies in an already stretched community. As noted by Uchitel, the overlaps between venues were significant enough that manuscripts could be sent almost anywhere. This led to the wasteful practice of "forum shopping," where authors of a rejected paper would send their work to a different venue with minimal changes, hoping for a better stochastic outcome. The effort in reviewing the original manuscript was completely wasted because:

- There was no incentive for transparency about a manuscript's history.
- Anonymous reviews were not shared between venues.
- Organizers had no access to reviewer names to request re-reviews.

While manuscripts jumped from venue to venue, reviewers did not. Resources were tied to specific venues, treating the global reviewer pool as fragmented silos.

To solve this collective-action failure, the top SE journals formed the **SE Journal Alliance** in 2026. To solve the collective-action failure of reviewer burnout, the top SE journals formed an informal alliance in 2026. This group established shared norms for reviewer load and, crucially, a portable credit layer.

This alliance shifted our work from competition to a collaborative stance, sharing procedures, infrastructure, reviewers, and historical data.

- **Portable Reviews:** In an accepted paper today, one can see the full lineage of its evolution across the Alliance. Previous reviews accompany the paper, preventing the loss of valid critiques and stopping the "reset" button of forum shopping.
- **Shared Credit & Norms:** The Alliance established shared norms for reviewer load and a "portable credit layer." This system tracks editorial and review contributions across all member venues, converting "invisible labor" into durable academic capital. By making this data visible to hiring and

tenure committees, the community transformed peer review from a volunteer burden into a recognized metric of professional standing.

In a sense, venues today are less "paper-evaluation-machines" and more like digests with specific viewpoints, selecting from a shared, community-wide pool of evaluated research.

2.2 The Lottery

Given the randomness reported above, it was natural to take the advice of Roumbanis [9] and restructure journal reviewing as a lottery process. Papers were scored by very fast moving pre-review teams (previously known as the "desk reject team"). The higher the score, the more likely a paper was selected for a traditional human review. All papers that scored over a certain threshold were always reviewed. Others took their chances with the lottery. By adjusting this threshold according to the number of available reviewers, journals could control how many reviews were needed.

Crucially, because the lottery reduced the total volume, we could afford to abandon the "one-shot" review model. It was replaced by a **continuous dialog process**. The "one-shot" review model, which often amplified misunderstanding and noise, was replaced by a continuous dialogue process. Every paper passing the initial gates triggers a private discussion page where authors, reviewers, and editors engage in structured Q&A. This interaction improves calibration and allows for the "meritorious acceleration" of high-potential work. Post-decision, these dialogues are published alongside the paper (with protected reviewer identities), serving as a vital educational resource and a record of the paper's evolution.

Another aspect that helped is a detailed pre-review process - as is common in journals like Nature - where only the most promising papers are sent out for peer review. TOSEM journal already started this process in 2025, with three levels of screening before a paper is sent for peer-revuew, by Editor in Chief, Senior Associate Editor and Associate Editor.

2.3 Review = Automatic + Manual

To manage this volume without surrendering quality, we adopted the "Human+AI" philosophy proposed by Manrai et al. [4]. They argued that AI should never be an arbiter of scientific merit, but serves as essential infrastructure.

We therefore moved away from "AI scoring" (which implies judgment) to "AI Triage" (which implies compliance). When reviewers look at a paper, they are checking for many items, some of which are purely structural and can be detected automatically.

Some automatic support proved useful for this triage process. The authors of this paper do not endorse fully automatic triage of academic papers. Nevertheless, when reviewers look at a paper, they are checking for many things, some of which can be detected automatically. For example:

- In the Ralph Empirical Standards initiative [8], SE researchers from around the world have commented on the features they expect in dozens of different kinds of SE papers.
- Our reading of those documents is that they contain approximately 135 review items. After asking Gemini to abstract those items, we found that there are variants of 12 core features. Significantly for our discussion here, some of these could be reviewed either by automatic methods or humans glancing very quickly at a paper.
- We hence proposed a hierarchical triage system where high-cost human review is only triggered if lower-level requirements are met.

By 2028, Open Science became a mandatory functional requirement to scale "Structural Gates," replacing static PDFs with a "Compilation Model." Submissions must now "compile"—executing analysis scripts in standardized containers—before human review, with exceptions only for justified sensitive industrial data. This shift enabled the AI revolution by allowing agents to automatically verify reproducibility and statistical consistency, freeing experts to focus entirely on interpretation, novelty, and high-level logic rather than broken technical details (i.e. where the underlying mechanics—data, code, or statistics—functionally fail or contain errors).

After that shift, here is our review plan.

Phase 1: Structural Gates (Automated) The entry level consists of binary checks. These are low-cost, structural validations. Failure here indicates an immature draft or policy violation.

- **Hard Exclusions:** Fail if clearly unrelated domain, not scientific manuscript, or violates alliance policy.
- **Structural Checks:** Presence of Related Work, Threats/Limitations, Results vs. Discussion distinction, and Replication Artifacts (code/data links).

Phase 2: Descriptive Gates (Extraction) This tier scans for definitions and risks. It creates a "Scope Profile" to simplify human review (Fast/Medium speed).

- **Scope Profile:** Extract claimed contribution type, objects of study, evaluation mode, and keywords/taxonomy mapping.
- **Risk Flags:** Flag mismatches to venue expectations or misalignment between claimed and AI-generated descriptions (non-failing).
- **Essential Definitions:** Verification of stated research goals, context, and variables.

Phase 3: Cognitive Review (Human Expert) Only papers passing prior gates reach experts. This phase focuses on logic, validity, and flag resolution (Slow/Deep speed).

- **Scope Adjudication:** Decide in/out of scope with rationale based on Phase 2 flags. May judge borderline/novel work requiring reframing.
- **Deep Review:** Assess motivation, sampling logic, methodology, and analysis. Double check the Related Work for discussion of relevant connections
- **Evaluation:** Rigor of validation. For experiments, standard metrics; for vision papers, self-critical addressal of counter-arguments.

While controversial, this triage method eliminated the incredibly time-consuming "hair-splitting" required to rank top-tier papers. It reduced the adversarial nature of reviews. However, it was only partially adopted. By 2030, many conferences used a lottery for the "middle tier" of papers, drastically reducing the debate time during PC discussions.

One unexpected benefit of the lottery was that it enabled a wider range of artifacts to be published. For example, GitHub repositories

with clear README files could be parsed and scored automatically, allowing review teams to evaluate well-motivated code with illustrative examples like any other research output.

2.4 Cathedrals and Bazaars

The structural changes described above would have failed without a parallel shift in the culture of incentives. By 2026, it was clear that the "one size fits all" publication model was forcing two distinct scientific species into the same cage.

To resolve the crisis, we formalized a "Two-Speed" culture, effectively recognizing that a healthy ecosystem needs both *Cathedrals* and *Bazaars* (otherwise known as “deep science” and "agile science”). For the foundational work, in deep science "Cathedral" papers[4] authors explored topics in extensive depth:

- **Deep PhD:** In cathedral science, the standard PhD thesis in Software Engineering shifted to include only one scientific publication. However, this single contribution was required to be of very high depth and rigor.
- **Hiring Metrics:** Hiring committees and tenure boards ceased "paper counting." Instead, they adopted the "Rule of Three": candidates were evaluated solely on the qualitative assessment of their three most significant publications. We note that certain countries already follow this practice.

This restored the "velocity of depth," allowing reviewers to scrutinize fewer submissions with significantly greater diligence.

Simultaneously, we stopped treating velocity as a vice. We acknowledged that "Bazaar" agile scientists (often highly practitioner-focused, exploring rapid tool developments and timely empirical snapshots) were being suffocated by the multi-year paper cycles. Conversely, when these rapid inputs were forced into the slow lane, they clogged the review machinery for everyone else. In SE, the effects of this clogging were painfully apparent. In a recent post, Luo [3] offers a heuristic ranking for how many people have to work together to write a paper in different fields. After normalizing the effort for machine learning papers to one, Luo ranks the effort associated with other fields (see Figure 2). Note that the author density required for SE papers was much higher than most other areas; i.e., there was ample space to redesign what we mean as a “paper”.

It was realized that "Bazaar" research needed a different validity construct entirely—one based on utility and immediacy rather than archival permanence. The specific mechanisms we engineered to support this fast stream are detailed in the next section.

2.5 Unbundling the Monolith

As shown in Figure 3, standard technical papers are a container of *motivation*, *method*, and *results*. But by 2030, we realized that forcing every contribution to carry all three containers was inefficient.

For the deep science track, we allowed researchers to publish modular units (Micro-publications):

- A **Vision Statement** only needs motivation.
- A **Registered Report** contains just motivation and method.
- A **Tools Paper** or **Literature Review** may be just method.
- A **Replication Paper** might be just motivation and results.

[4]Adapted from https://compsci.science/slowscience.

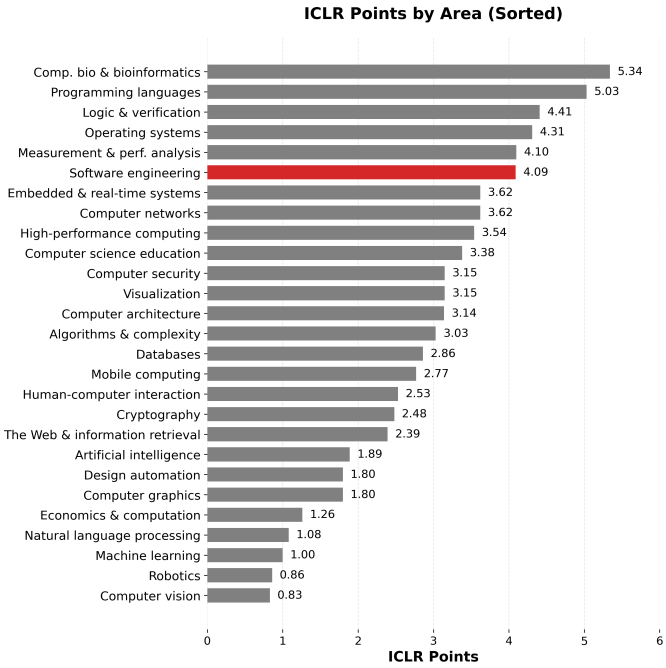

Figure 2: SE author density is four times that of machine learning (from [3]) reflecting either higher effort is required or vastly different authorship cultures.

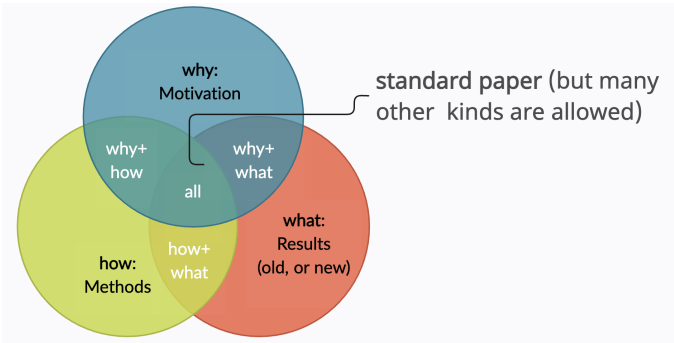

Figure 3: Different kinds of papers.

Prior to 2025, the SE community was already experimenting with dividing up papers in this way. For example, Ernst and Baldassarre [2] described *registered reports* as scientific publications that begin with the peer review of a detailed research protocol before the study is conducted. In terms of Figure 3, the protocol would be a publication of motivation and methods, and the subsequent results report would be mostly results.

Unbundling the "monolithic paper" into these smaller units allowed us to specialize reviewing. For example, a "Methodology" paper could be reviewed quickly by a statistics expert, without needing to evaluate a year’s worth of results simultaneously. By 2030, this reduced the average reading time per reviewer by 40%,

as they only had to critique one aspect of the research life cycle at a time.

For the faster agile track, we solved the "ivory tower" problem by detaching the practitioner payload from the scientific manuscript. Journal papers are no longer considered by software practitioners as "write-only," inaccessible works. Instead, every accepted submission is now accompanied by AI-generated Summaries and Videos.

- **Access:** This content is placed outside the "paywalls" of traditional publishers. It requires no subscription, allowing it to reach the global engineering workforce instantly.
- **Format:** These artifacts explain the context, problem, and key findings in a way that is accessible and time-efficient. Practitioners can digest the core contribution in minutes.
- **Integration:** Artifacts like tools or datasets are embedded directly into practitioner workflows via AI tooling supports.

Crucially, these summaries are *not* subject to pre-publication peer review. They are published independently. The quality control here is not "guarding," but usage. Practitioner feedback on these artifacts is tracked and—vital to the culture shift mentioned in Section 4—this feedback now counts towards meeting impact criteria in academic career paths.

2.6 Escaping the Benchmark Graveyard

By 2025, empirical software engineering faced a paradox. We had more data than ever, but less insight. The root cause was our relationship with benchmarks.

The 2026 "Moot" paper [5] warned that benchmarks can be a *catalyst* or a *cage*. Used thoughtfully, they broaden the questions we ask. But used carelessly, they suffer a tragic life cycle:

(1) **Rejected:** "Find Data? That will never happen."
(2) **Respected:** "Fine... it helps sometimes."
(3) **Expected:** "You must compare to this baseline."
(4) **Exhausted:** The research graveyard where innovation stalls; results become repetitive and derivative.

By the mid-20s, several SE sub-fields had entered the "Exhausted" phase. The widespread reuse of open datasets had reduced validation to a narrow form of testing. Researchers were merely demonstrating that "the method works on the existing dataset" or "the tool is better by x% than another tool."

This practice occurred without sufficient critical examination of the validity of the chosen dataset itself. We were optimizing for the cage, ignoring whether the method was effective in real-world settings or industrial environments. Consequently, impact stagnated—industry looked to non-peer-reviewed blogs because academic papers were winning games on stale scoreboards.

To escape the graveyard, the Journal Alliance introduced the "Catalyst Criteria" in 2027. We explicitly stopped rewarding papers that only offered minor percentage gains over existing baselines. Instead, acceptance now requires work that actively expands the field, such as:

- **New Tasks or Domains:** Moving beyond the "standard" datasets to prove utility in messy, real-world contexts.
- **Challenging Assumptions:** Using benchmarks to disprove prevailing wisdom rather than just climbing a leaderboard.
- **Extending the Benchmark:** Contributing back to the community by improving the data, tooling, or organization of the benchmark itself.

This shift transformed benchmarks from a ceiling that limited us into a floor upon which we stood to reach higher.

3 Conclusion

Looking back from 2036, the crisis of the mid-20s appears not as a disaster, but as a necessary correction. The "bureaucratic anomaly" of 2025—where we tried to apply 19th-century reviewing methods to 21st-century AI velocity—forced us to innovate.

The solution lay in humility. By admitting that peer review is stochastic, we adopted the Lottery. By admitting that PDFs are opaque, we demanded Executable Papers. By admitting that practitioners ignore paywalls, we built the Bazaar.

Ultimately, the Journal Alliance didn't just save our venues; it saved our time. We moved from a system of "Quality Control by Guarding" (where we spent 90% of our energy filtering noise) to "Quality Control by Dialog" (where we spend 90% of our energy improving signal). Software Engineering is no longer just a publishing machine; it is, once again, a scientific community.

Disclaimer

This paper is co-authored by several editor-in-chiefs. The views expressed capture their individual experiences in the role, and do not necessarily capture the views of the professional societies for their corresponding journals.

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